

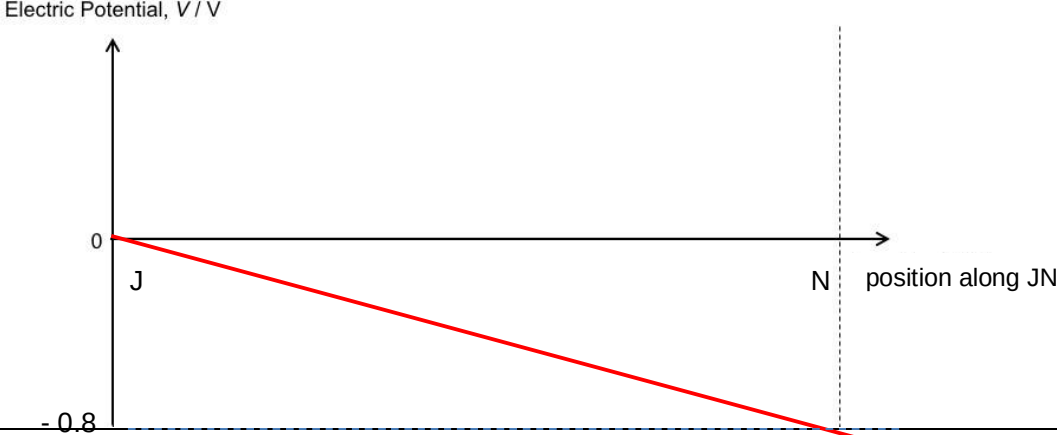
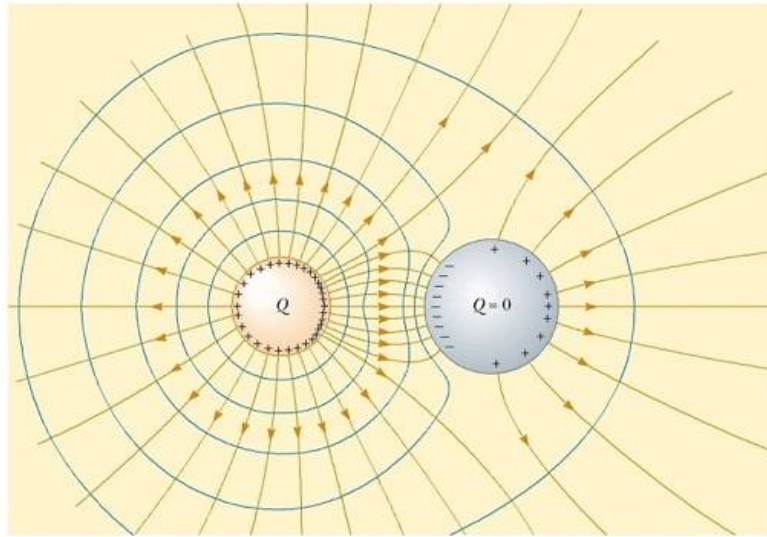
8	(a)	(i)	It is the energy transferred per unit charge from other forms into electrical energy when charge is moved round a complete circuit.	
		(ii)	Energy supplied by the cell = energy dissipated in the external and internal resistance Energy supplied by the cell per unit charge = energy dissipated in the external and internal e.m.f. of cell per unit charge = p.d across the external circuit + p.d across the internal resistance of the cell $E = V + V_r$ $E = V + Ir$ $V = E - Ir$	
	(b)	(i)	$V = E - Ir$ At (300, 0.4), $0.4 = E - (300 \times 10^{-3})r$ --- (1) At (100, 1.2), $1.2 = E - (100 \times 10^{-3})r$ --- (2) (2) – (1): $0.8 = (200 \times 10^{-3})r$ $r = 4 \Omega$ Sub into (2): $1.2 = E - (100 \times 10^{-3})(4)$ $E = 1.6 \text{ V}$	
		(ii)	The emf is 1.6 V as calculated in b(i). For maximum power to be delivered to JN, the resistance of JN must be the same as that of the internal resistance. Hence the potential difference across the internal resistance must be the same as that across JN. Since JN is the only external resistor in the circuit, the potential difference must be 0.8 V (which is 50% of the emf), as it is a series connection. Since J is connected to earth, potential at J is <u>0 V</u> and since pd drops from J to N, N is at a lower potential of <u>-0.8 V</u>. (allow for ecf)	
		(iii)	Electric Potential, V / V  position along JN	

			Fig. 8.3 straight line graph starting with 0 V at J and ending at – 8 V at N Reasons: Current along JN is constant Potential drops along wire JN is equal to current x resistance of wire Resistance of wire is proportional to the length since wire is uniform Potential difference drop is proportional to length of wire measured from J, therefore graph is linear.	
		(iv)	As the battery is used, its <u>e.m.f. is reduced</u> but its <u>internal resistance will be increased</u>. For maximum power transfer, the external resistance should be larger now since it is equal to that of the internal resistance of the battery, hence the wire JN will be longer, so that jockey J has to be shifted to the left.	
	(c)	(i)	It is the region of space where a charged particle experiences an electric force.	
		(ii)1	approximately equal distribution of positive charges electric field pattern must correspond to the position of the charges, correct directional arrows indicated	

2.



This solution also shows the equipotential lines.

closer distribution of positive charges on A facing B
equal number of positive charges and negative charges on B, at the correct sides
correct direction of electric field lines for A and B
correct shape of the electric field pattern

Note: equipotential lines are not required

End of Solutions